Experiment - 1 Date: 20/07/2026

Aim: Study of various network commands used in Linux and windows

=> Windows Commands

i) arp -a: Short form for address resolution protocol. Shows IP address.

Output:

(172.16.8.127) at 00:27:0e:13:b7:8c [ether] on enp3s0
(172.16.8.149) at 00:27:0e:13:b3:be [ether] on enp3s0
(172.16.8.212) at 7c:57:58:38:da:a3 [ether] on enp3s0
(172.16.8.186) at 00:27:0e:13:b5:77 [ether] on enp3s0
(172.16.11.208) at 7c:57:58:38:46:87 [ether] on enp3s0

ii) Hostname: Simplest of all TCP/IP commands. Displays name of the computer.

Output:

localhost.localdomain.

iii) ipconfig/all: Displays detailed config. info about TCP/IP connection

Output:

1: lo: inet 127.0.0.1/8 scope host lo
2: enp3s0 inet 172.16.8.127/22

iv) ping
nbtstat :- Helps solve problems with NetBIOS name resolution. Packet Internet Groper commands.

Output

PING localhost (localhost (::1)) 56 data bytes
64 bytes from localhost (::1): icmp_seq=1 ttl=64 time=0.012ms
64 bytes from localhost (::1): icmp_seq=2 ttl=64 time=0.043ms

(v) nslookup : A tool used to perform DNS lookups on Linux.

Output:

Server: 172.16.8.1
Address: 172.16.8.1#53

Non-authoritative answer:
Name: www.google.com
Address: 142.251.157.119

Name: www.google.com
Address: 142.251.156.119

(vi) Pathping : Combination of ping and tracert commands.

output: Over a max of 30 hops

0 suji [172.16.97.153]
1 suji [172.16.97.153]

Computing statistics for 125 seconds

Hop	RTT	Source to here. This node/link	Address
		host/sent=lost/sent=lost	
0	~~0ms~~	0/100=0%	0/100=0% 172.16.97.153
1	0ms	0/100=0%	0/100=0% 172.16.8.1

Trace Complete.

Linux Commands:

(i) ifconfig: Is a staple in many sysadmin's tool belt for configuring / trouble shooting networks.

Output:

enp3s0: flags = 4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 150
inet 172.16.8.12 netmask 255.255.252.0 broadcast 172.16.11.255
inet6 fe80::abbb::cdb1::131c::6e50 prefixlen 64 scopeid 0x2
ether 00:27:0e:13:bb:d9 txqueuelen 1000 (Ethernet).

(ii) mtr: Serves as a network diagnostic tool

Output:

mtr google.com

local host. local domain()

Host	Packets						
	loss%	snt	last	Avg	Best	Wrst	stdev
	0.0%	17.8	0.1	0.1	0.0	0%	0.0

(iii) tcpdump - Designed to capture and display packets

Output:

tcpdump -D

1. enp3s0 (up, Running)
2. any (Pseudo: device that captures all interfaces
3. lo (up, Running, loopback)

4: wlp3s0

5. Bluetooth 0: (Bluetooth adapter)

Result :-

Successfully executed Linux and Windows commands.